

RED TEAM PROJECT SCOPE

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INTRODUCTION TO RED TEAMING

1. Penetration Testing

The primary objective of a penetration test is to identify and exploit vulnerabilities in a target system in order to measure the risk associated with the exploitation of the target attack surface.

Penetration testing provides an organization with a much more accurate understanding of how an attacker could gain access to their environment and provides valuable information on where detection and defense capabilities need to be improved.

Penetration testing is quite limited with regards to its ability to emulate or simulate a real threat actor, mainly because of the scope. Penetration tests can be either black box or white box and can be very loud.

Given that the goal of a penetration test is to identify and exploit vulnerabilities in target systems, risk is typically measured and is limited to the workstations or assets within the predefined scope and does not encompass employees, defenders, and processes.

2. Limitation Of Penetration Tests

Given the underlying objective, penetration testers usually do not have to worry about triggering alerts or evading detection. As a result, penetration testers are typically noisy and loud, while real-world threats are not.

A traditional penetration test is likely to ignore attack vectors like social engineering and physical intrusion (which are very common with real adversaries).

Penetration testers must abide by strict rules of engagement and the predefined scope.

In certain cases, the organization may whitelist the penetration tester's attack infrastructure, and in some cases, defense and detection mechanisms may be relaxed.

3. Why Red Team

Red teaming allows you to assess and measure the effectiveness and resilience of employees, defenders, and processes used to defend a company's digital infrastructure.

Very useful for measuring the Blue Team's ability to detect and defend against adversaries.

Used to train the Blue Team. Defenders require frequent training and practice in order to be effective.

Provides exposure to real-world threats/APTs and their corresponding TTPs, tradecraft, and malware.

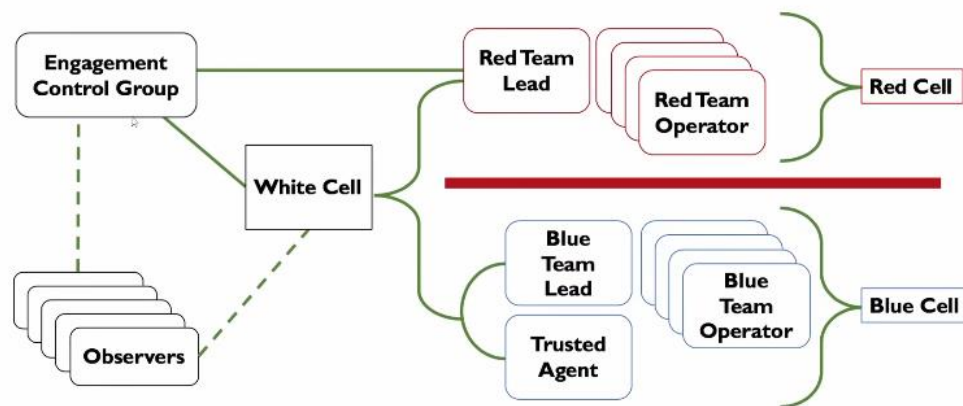
4. Essential Terminology

TERM	DEFINITION
TTPs	Tactics, Techniques & Procedures.
Tradecraft	Techniques and procedures during an attack campaign.
OPLOG	Operator logs are the records generated by Red Team operators during an engagement. These logs have specific and required fields that must be captured.
C2	Command and Control
Exfiltration	Process of extraction of information/data from a target system through a covert channel.
IoC	Indicator of compromise – Artifacts used to identify adversarial activity.
OPSEC	Operational Security – What the Blue Team can observe and is used to minimize exposure.
Operational Impact	Effect of an objective driven action within a target environment.
Situational Awareness	Phase of a Red Team operation used to gather information on targets and the target environment. This information used to determine the next action.
CTI	Cyber Threat Intelligence – Information collected, aggregated, analyzed and interpreted to provide the context for decision-making processes regarding threats.

5. Types Of Red Team Engagements

ENGAGEMENT	USE CASE/DEFINITION
Full Simulation	Simulates a threat/adversary's attack flow.
Adversary Emulation	Emulating/Mimicking an adversary's/APT's TTPs with little or no deviation.
Assumed Breach	The Assumed Breach Model assumes a threat has some level of access to a target at the initiation of the engagement.
Table-top Exercise	An over the table simulation where scenarios are discussed between the red and blue teams to evaluate how they would theoretically respond to certain threats.

6. Red Team Roles And Responsibilities



ROLE	RESPONSIBILITY
Red Cell	Group that plays OPFOR (opposing force) during red vs. blue exercises. A red cell is the components that make up the offensive portion of a red team engagement that simulates the strategic and tactical responses of a given target. The red cell is typically comprised of red team leads and operators and is commonly referred to as Red Team instead of Red Cell.
Blue Cell	The blue cell is the opposite side of red. Is it all the components defending a target network. The blue cell is typically comprised of blue team members, defenders, internal staff, and an organization's management.
White Cell	Serves as referee between Red Team activities and defender responses during an engagement. Controls the engagement environment/network. Monitors adherence to the ROE.

ROLE	RESPONSIBILITY
Red Team Lead	Serves as the operational and administrative lead for the Red Team. Conducts engagement, budget, and resource management for the Red Team, Provides oversight and guidance for engagements, capabilities, and technologies. Ensures adherence to all laws, regulations, policies, and Rules of Engagement.
Red Team Assistant Lead	Assists the team lead in overseeing engagement operations and operators. Can also assist in writing engagement plans and documentation if needed.
Red Team Operator	Complies with all Red Team requirements under the direction of the Red Team Lead. Operational executor of the engagement. Applies Red Team TTPs to the engagement. Provides technical research and capability to the Red Team. Keeps detailed logs during each phase of the engagement.